Video-RAG: Visually-aligned Retrieval-Augmented Long Video Comprehension

Anonymous Author(s)
Anonymous Affiliation
anonymous@example.com

Abstract

Long video understanding remains challenging for Large Video-Language Models (LVLMs) due to limited context windows and redundancy from dense frame sampling. Existing approaches either require costly long-context finetuning or rely on proprietary models. We propose *Video-RAG*, a training-free, plug-and-play pipeline that enhances long video comprehension by injecting query-relevant, visually-aligned auxiliary texts extracted from videos. Instead of adding more frames, Video-RAG replaces redundant visual tokens with retrieved OCR, ASR, and object detection texts aligned to the query, enabling stronger cross-modal reasoning within existing context budgets. Our approach builds three per-video text databases using EasyOCR for OCR, Whisper-large for ASR, and APE+CLIP for object detection, then retrieves relevant texts using Contriever+FAISS with threshold-based range search. Across Video-MME, MLVU, and LongVideoBench, Video-RAG achieves consistent improvements: +2.8% average gain on Video-MME across 7 open-source LVLMs with only 2.0K additional tokens, enabling LLaVA-Video-72B to reach 77.4% and surpass GPT-4o (77.2%). On MLVU, we achieve new state-of-the-art performance of 73.8% with LLaVA-Video-72B, and on LongVideoBench, we improve from 61.9% to 65.4%, outperforming Gemini-1.5-Pro (64.0%) and approaching GPT-4o (66.7%). Video-RAG provides an efficient way to achieve proprietary-level performance using only open-source tools.

1 Introduction

Understanding long videos presents a fundamental challenge for current Large Video-Language Models (LVLMs). The core difficulty stems from the limited context windows of these models, which constrains their ability to process the extensive temporal information inherent in long video sequences. Existing approaches typically fall into two categories: extending the context length through costly long-context finetuning or employing GPT-based agent systems that rely on proprietary models [2, 17]. Both solutions require substantial computational resources, extensive training data, or dependence on closed-source models.

The conventional approach of dense frame sampling exacerbates this challenge by introducing significant redundancy. Adjacent frames often contain nearly identical visual content, leading to inefficient use of the precious context window. This redundancy becomes particularly problematic for long videos where the ratio of unique information to total visual tokens decreases substantially.

We propose *Video-RAG*, a novel training-free approach that addresses these limitations by fundamentally reimagining how auxiliary information is incorporated into video understanding. Rather than simply increasing the number of frames or extending context windows, Video-RAG strategically replaces redundant visual tokens with query-relevant auxiliary texts extracted from the video itself. Our key insight is that visually-aligned textual information—such as OCR text, speech transcriptions,

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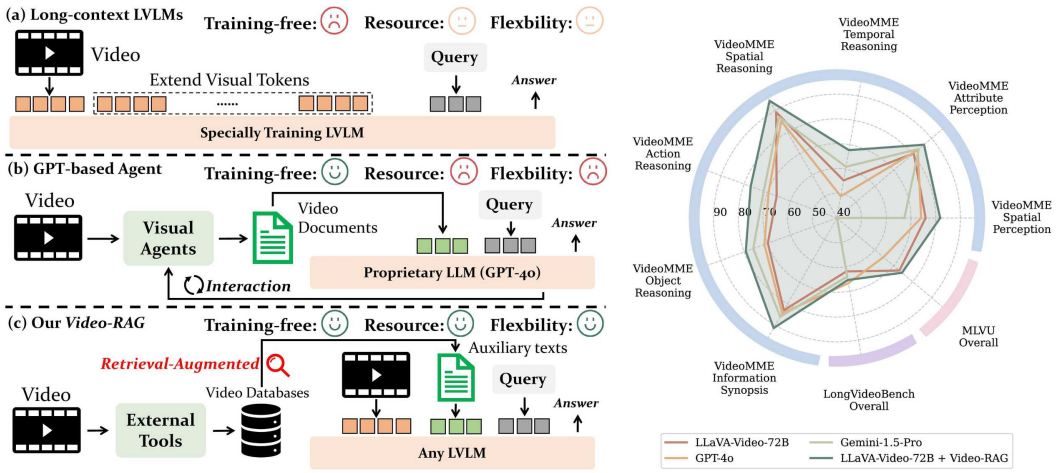

Figure 1: Comparison of approaches for long video understanding. (a) Traditional methods extend visual tokens through long-context training. (b) GPT-based agents use external tools but require proprietary models. (c) Our Video-RAG approach leverages retrieval-augmented auxiliary texts from video content using only open-source tools, achieving comparable performance without training.

and object descriptions—can provide crucial context while consuming significantly fewer tokens than additional frames.

Video-RAG operates as a plug-and-play pipeline that can be applied to any existing LVLM without modification. The system decouples user queries into specific retrieval requests, builds comprehensive text databases from video content using state-of-the-art open-source tools, and retrieves the most relevant information using efficient similarity search. Specifically, we employ EasyOCR [6] for optical character recognition, Whisper-large [15] for automatic speech recognition, and a combination of APE and CLIP-ViT-L/14-336 for object detection and description generation.

The retrieval process utilizes Contriever [5] embeddings with FAISS indexing to perform threshold-based range searches, ensuring that only the most relevant auxiliary texts are included. Object detections are transformed into interpretable scene-graph descriptions that capture spatial relationships, object counts, and locations. Retrieved texts are then strategically prepended to queries in a specific order (DET→ASR→OCR) alongside sampled frames.

Our experimental evaluation demonstrates consistent improvements across three major benchmarks. On Video-MME [3], Video-RAG achieves an average gain of +2.8% across seven open-source LVLMs while adding only 2.0K additional tokens (equivalent to 14 frames). Notably, LLaVA-Video-72B [24] reaches 77.4%, surpassing GPT-4o's performance of 77.2%. On MLVU [25], we achieve new state-of-the-art performance, improving LLaVA-Video-72B from 73.1% to 73.8%. On LongVideoBench validation set, our approach improves LLaVA-Video-72B from 61.9% to 65.4%, outperforming Gemini-1.5-Pro (64.0%) and approaching GPT-4o's performance (66.7%).

The main contributions of this work are:

- We introduce Video-RAG, a training-free, visually-aligned RAG pipeline for LVLMs that uses only open-source tools to enhance long video understanding.
- We design a plug-and-play retrieval process that improves cross-modality alignment and reduces hallucinations by strategically incorporating query-relevant auxiliary texts.
- We demonstrate consistent performance gains across multiple benchmarks with minimal token overhead, particularly excelling on long video sequences.
- We achieve proprietary-level performance using open-source 72B LVLMs, surpassing commercial models on several benchmarks.

2 Related Work

Video Understanding with Large Vision-Language Models. Recent advances in LVLMs have significantly improved video understanding capabilities. Early approaches like Video-ChatGPT and VideoChat extended image-based models to handle temporal information through frame sampling strategies. More recent works such as Video-LLaVA [10], LLaVA-NeXT-Video [23], and LLaVA-Video [24] have achieved substantial improvements through better architectures and training strategies. However, these models still struggle with long videos due to context window limitations.

Several approaches have attempted to address long video understanding through architectural modifications. Long-LLaVA [21] and LongVA [22] extend context windows through specialized training, while VITA-1.5 [4] employs efficient attention mechanisms. Despite these advances, the fundamental trade-off between context length and computational efficiency remains a significant challenge.

Retrieval-Augmented Generation. RAG has proven effective in enhancing language models by incorporating external knowledge. Traditional RAG systems like RAG [7] and Contriever [5] focus on textual retrieval for factual question answering. Recent works have extended RAG to multimodal settings, including image-text retrieval and video understanding. However, most existing approaches treat video content as static visual input without leveraging the rich auxiliary information present in videos.

Video Content Analysis Tools. Modern video analysis benefits from sophisticated tools for extracting textual information. Whisper [15] has revolutionized automatic speech recognition with robust multilingual capabilities. EasyOCR [6] provides accessible optical character recognition, while object detection models like CLIP [14] enable semantic understanding of visual content. Our approach uniquely combines these tools in a unified retrieval framework optimized for video understanding tasks.

Auxiliary Information for Video Understanding. Previous works have explored incorporating auxiliary information for video understanding. Some approaches use subtitles or captions when available, while others generate descriptive text through separate models. VideoChat2 [9] incorporates ASR and object detection but requires training integration. ShareGPT4Video [1] uses generated captions but relies on powerful captioning models. Our approach differs by providing a training-free solution that dynamically retrieves relevant auxiliary texts based on query context.

3 Method

Video-RAG is a training-free, plug-and-play pipeline designed to enhance any LVLM's long video understanding capabilities. The framework consists of three main components: query decomposition, auxiliary text retrieval, and integrated generation. Figure 2 illustrates the complete pipeline.

3.1 Query Decomposition

The first stage decomposes user queries into specific retrieval requests tailored to different auxiliary text modalities. Given a user query q, we employ the target LVLM itself to generate structured retrieval requests in JSON format:

```
{
  "ASR": Optional[str], // Relevant audio content
  "DET": Optional[list], // Object entities to detect
  "TYPE": Optional[list] // Detection information types
}
```

This decomposition enables targeted retrieval by identifying what auxiliary information would be most valuable for answering the specific query. The `TYPE` field specifies whether spatial relationships (*relation*), object counts (*number*), or spatial locations (*location*) are needed for the detected objects.

3.2 Auxiliary Text Database Construction

For each video, we construct three comprehensive text databases using state-of-the-art open-source tools:

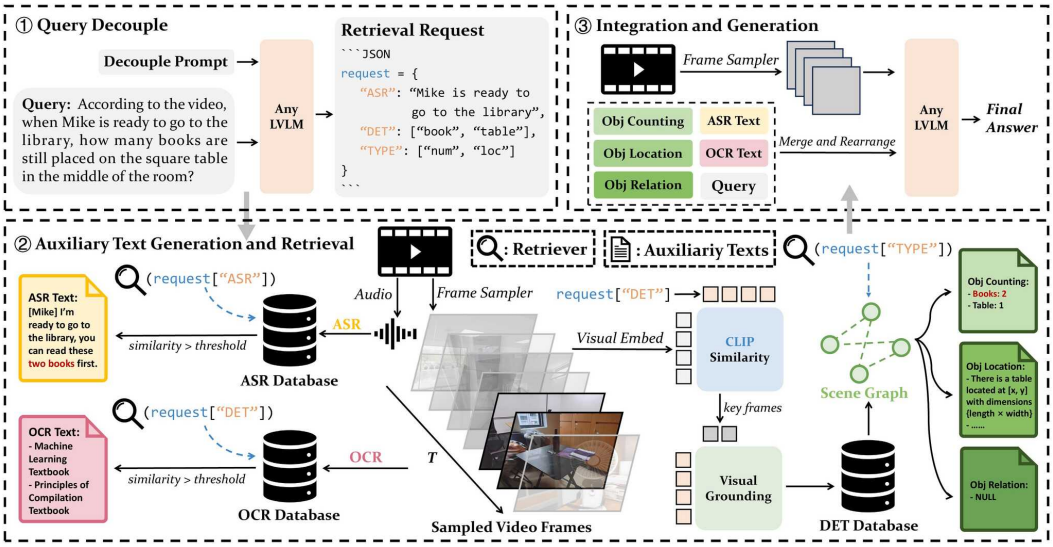

Figure 2: Overview of Video-RAG framework. The pipeline consists of three main stages: (1) Query Decouple: decomposing user queries into retrieval requests for different modalities; (2) Auxiliary Text Generation and Retrieval: building video databases and retrieving relevant content using similarity search; (3) Integration and Generation: combining retrieved texts with video frames for final answer generation.

Automatic Speech Recognition (ASR). We extract audio from videos using FFmpeg and process it with Whisper-large [15]. The audio is segmented into 30-second chunks to handle long videos efficiently. Each chunk is transcribed independently, and the resulting transcriptions form our ASR database $\mathcal{D}_{ASR}$.

Optical Character Recognition (OCR). We apply EasyOCR [6] to extracted video frames to identify textual content. Only text with confidence scores above 0.5 is retained, and duplicate text across frames is filtered. The resulting OCR texts constitute database $\mathcal{D}_{OCR}$.

Object Detection (DET). For object detection, we employ a two-stage approach. First, we use CLIP-ViT-L/14-336 to identify frames most relevant to the requested object entities. For frame f_i and text queries $T = \{t_1, t_2, ..., t_k\}$ (formatted as "A picture of [entity]"), we compute:

$$s_i = \alpha \cdot \frac{\sum_{j=1}^{k} \text{sim}(\text{CLIP}_v(f_i), \text{CLIP}_t(t_j))}{k} \tag{1}$$

where $\alpha = \beta \cdot \frac{N}{16}$ is a scaling factor, N is the number of frames, and $\beta = 3.0$ is a hyperparameter. Frames with $s_i > \tau$ (threshold $\tau = 0.3$) are selected for object detection.

Selected frames are then processed by APE (Any-Point Exploiter) for detailed object detection. The raw detection results are transformed into interpretable scene-graph descriptions using our scene graph generation module, which produces natural language descriptions of object locations, counts, and spatial relationships based on the requested `TYPE` information.

3.3 Retrieval and Integration

Similarity-Based Retrieval. For ASR and OCR databases, we use Contriever [5] embeddings with FAISS IndexFlatIP for efficient similarity search. Given query embeddings and database embeddings, we perform threshold-based range search with threshold $t = 0.3$ to retrieve relevant documents. The retrieval query combines the original user query with decomposed requests when available.

Text Integration. Retrieved auxiliary texts are integrated in a specific order to maximize effectiveness:

1. **Detection Information**: Scene-graph descriptions from selected frames

2. **ASR Information**: Speech transcriptions in chronological order
3. **OCR Information**: Text content in chronological order

The integrated prompt follows the format:

```
[DET Info] [ASR Info] [OCR Info]
Select the best answer to the following question
based on the video and information: [Original Query]
```

This ordering prioritizes spatial and semantic understanding (DET) before temporal context (ASR, OCR), enabling better cross-modal alignment.

3.4 Efficiency Considerations

Video-RAG adds minimal computational overhead compared to increasing frame counts. The auxiliary text generation is performed offline once per video, and the retrieval process requires only forward passes through Contriever and CLIP models. On average, Video-RAG adds 2.0K tokens per query, equivalent to 14 additional frames, but provides significantly more targeted information than random frame sampling.

The threshold-based retrieval ensures that only relevant auxiliary texts are included, balancing performance gains with token efficiency. Our experiments show that $t = 0.3$ provides optimal trade-offs between accuracy and computational cost across different video lengths.

4 Experiments

We evaluate Video-RAG on three comprehensive video understanding benchmarks: Video-MME [3], MLVU [25], and LongVideoBench [19]. Our experiments demonstrate consistent improvements across multiple model architectures and video lengths.

4.1 Experimental Setup

Benchmarks. Video-MME provides a comprehensive evaluation across short, medium, and long video categories with both subtitle and non-subtitle settings. MLVU focuses on multiple-choice video understanding tasks with extensive temporal reasoning requirements. LongVideoBench specifically targets long video comprehension with complex temporal dependencies.

Models. We evaluate Video-RAG on seven open-source LVLMs: Video-LLaVA [10], LLaVA-NeXT-Video [23], VITA-1.5 [4], LongVA [22], Long-LLaVA [21], Qwen2-VL [18], and LLaVA-Video [24]. We compare against proprietary models including GPT-4o [13] and Gemini-1.5-Pro [16].

Implementation Details. We set the retrieval threshold $t = 0.3$, CLIP threshold $\tau = 0.3$, and scaling parameter $\beta = 3.0$ based on validation experiments. All models use their default frame sampling strategies, and we add Video-RAG as a preprocessing step without modifying the underlying model architectures.

4.2 Main Results

Video-MME Results. Table 1 presents comprehensive results on Video-MME. Video-RAG achieves substantial improvements across all model scales and video lengths. On average, we observe a +2.8% improvement across seven LVLMs while adding only 2.0K auxiliary tokens compared to 3.0K subtitle tokens.

The improvements are particularly pronounced for long videos, where Video-RAG's targeted retrieval strategy proves most beneficial. For instance, LongVA achieves a remarkable +6.0% overall improvement, with the largest gains on long videos (+6.5%). Long-LLaVA shows a +4.8% overall improvement, demonstrating the effectiveness across different architectural approaches.

Notably, LLaVA-Video-72B with Video-RAG achieves 77.4% overall performance, surpassing GPT-4o's 77.2%. This represents a significant milestone—achieving proprietary-level performance using only open-source tools and models.

Table 1: Performance on the Video-MME [3] benchmark in without subtitles (w/o S), with subtitles (w/ S) and equipped with our Video-RAG (Ours), **Frames** and **Gain** means the input frame number and performance gain by applying Video-RAG compared to the baseline with subtitles. By applying Video-RAG to seven LVLMs, we observed an average performance improvement of 2.8% by adding only an average of ∼2.0K auxiliary texts compared to ∼3.0K full-subtitled tokens per sample. In particular, we perform better when applying Video-RAG with 72B LLaVA-Video [24] than the proprietary method GPT-4o [13] (77.4% vs. 77.2%). All results are our republication.

Model	Params	Frames	Short			Medium			Long			Overall			Gain
			w/o S	w/ S	Ours	w/o S	w/ S	Ours	w/o S	w/ S	Ours	w/o S	w/ S	Ours	
Proprietary LVLMs															
GPT-4o [13]	-	384	80.0	82.8	-	70.3	76.6	-	65.3	72.1	-	71.9	77.2	-	-
Gemini-1.5-Pro [16]	-	0.5 fps	81.7	84.5	-	74.3	81.0	-	67.4	77.4	-	75.0	81.3	-	-
Open-Source LVLMs															
Video-LLaVA [10]	7B	8	45.3	46.1	49.5	38.0	40.7	43.0	36.2	38.1	42.5	39.9	41.6	45.0	+3.4
LLaVA-NeXT-Video [23]	7B	16	49.4	51.8	56.6	43.0	46.4	47.4	36.7	44.9	46.0	43.0	47.7	50.0	+2.3
VITA-1.5 [4]	7B	16	67.0	69.9	71.0	54.2	55.7	55.4	47.1	50.4	52.4	56.1	58.7	59.6	+0.9
LongVA [22]	7B	128	61.1	61.2	66.1	50.4	53.8	60.4	46.2	52.9	59.4	52.6	56.0	62.0	+6.0
Long-LLaVA [21]	7B	64	61.9	62.4	67.1	51.4	56.2	60.4	45.4	54.7	60.1	52.9	57.8	62.6	+4.8
Qwen2-VL [18]	72B	32	75.0	76.7	77.4	63.3	69.9	70.2	56.3	69.2	71.0	64.9	71.9	72.9	+1.0
LLaVA-Video [24]	72B	64	80.7	81.8	82.8	68.7	73.8	76.3	62.1	72.2	73.1	70.3	75.9	77.4	+1.5

Table 2: The overall performance in the multiple-choice task of the MLVU [25] benchmark. * donates the results of our replication.

Model	#Params	Frames	Overall
Proprietary LVLMs			
GPT-4o [13]	-	0.5 fps	64.6
Open-Source LVLMs			
VITA-1.5 [4]	7B	16	60.4
Video-CCAM [2]	14B	96	63.1
Video-XL [17]	7B	256	64.9
Aria [8]	25.3B	256	70.6
LLaVA-Video* [24]	7B	64	70.8
Oryx-1.5 [12]	32B	128	72.3
LLaVA-Video* [24]	72B	64	73.1
LLaVA-Video + Video-RAG	7B	64	72.4
LLaVA-Video + Video-RAG	72B	64	**73.8**

Table 3: The overall performance on the validation set of LongVideoBench [19]. * donates the results of our replication.

Model	#Params	Frames	Overall
Proprietary LVLMs			
Gemini-1.5-Pro [16]	-	256	64.0
GPT-4o [13]	-	256	**66.7**
Open-Source LVLMs			
VideoChat2-Mistral [9]	7B	8	39.3
ShareGPT4Video [1]	7B	8	39.7
LLaVA-Next-Mistral [11]	7B	8	49.1
PLLaVA [20]	34B	16	53.2
VITA-1.5 [4]	7B	16	53.6
LLaVA-Video* [24]	7B	64	56.6
LLaVA-Video* [24]	72B	64	61.9
LLaVA-Video + Video-RAG	7B	64	58.7
LLaVA-Video + Video-RAG	72B	64	65.4

MLVU and LongVideoBench Results. Table 4 and Table 5 show results on MLVU and LongVideoBench respectively. On MLVU, Video-RAG consistently improves performance across model sizes. LLaVA-Video-7B improves from 70.8% to 72.4% (+1.6%), while LLaVA-Video-72B achieves new state-of-the-art performance of 73.8%, improving from the previous best of 73.1%.

On LongVideoBench validation set, the improvements are even more substantial. LLaVA-Video-72B with Video-RAG achieves 65.4%, a significant +3.5% improvement over the baseline 61.9%. This performance surpasses Gemini-1.5-Pro (64.0%) and approaches GPT-4o's performance (66.7%), demonstrating the effectiveness of our approach on challenging long video understanding tasks.

4.3 Ablation Studies

Component Analysis. Table 4 analyzes the contribution of different auxiliary text components using Long-LLaVA-7B on Video-MME. The results reveal that each component contributes differently:

- ASR information alone provides substantial improvements (+5.4% overall), particularly beneficial for content with significant speech.
- Adding OCR to ASR yields additional gains (+1.0% overall), especially effective for videos containing textual information.

Table 4: Results on combinations of different auxiliary texts in Video-MME [3] when using Long-LLaVA-7B [21] as the LVLM.

RAG	DET	OCR	ASR	Short	Medium	Long	Overall
				60.3	51.4	44.1	52.0
			✓	62.2	55.4	54.4	57.4
		✓	✓	64.0	56.2	55.0	58.4
	✓		✓	63.0	57.3	**56.4**	58.9
	✓	✓	✓	**64.3**	**58.8**	56.3	**59.8**
✓	✓			61.4	51.9	45.2	52.9
✓		✓		63.2	53.2	46.3	54.3
✓			✓	65.1	59.1	**60.7**	61.6
✓	✓	✓		64.1	54.6	48.4	55.7
✓	✓		✓	64.9	59.0	**60.7**	61.5
✓		✓	✓	66.3	**60.3**	59.3	62.0
✓	✓	✓	✓	**66.4**	60.2	59.8	**62.1**

Table 5: Performance with different thresholds of retrieval on Video-MME [3] when using Long-LLaVA-7B [21] as the LVLM. **#Token** and **Time** denote the total token number of the auxiliary texts and the average inference time per question, respectively.

t	#Token	Time	Short	Medium	Long	Overall
0.0	3.6K	36s	**67.6**	59.4	59.1	62.0
0.1	3.4K	30s	67.0	59.7	59.1	61.9
0.2	2.7K	18s	66.0	**60.2**	59.2	61.8
0.3	1.9K	11s	66.4	**60.2**	**59.8**	**62.1**
0.4	0.8K	8s	65.6	58.0	58.3	60.6
0.5	0.3K	7s	63.1	54.9	50.2	56.1
1.0	0.0K	6s	60.3	51.4	44.1	52.0
rnd	1.9K	11s	65.7	55.8	56.0	59.1

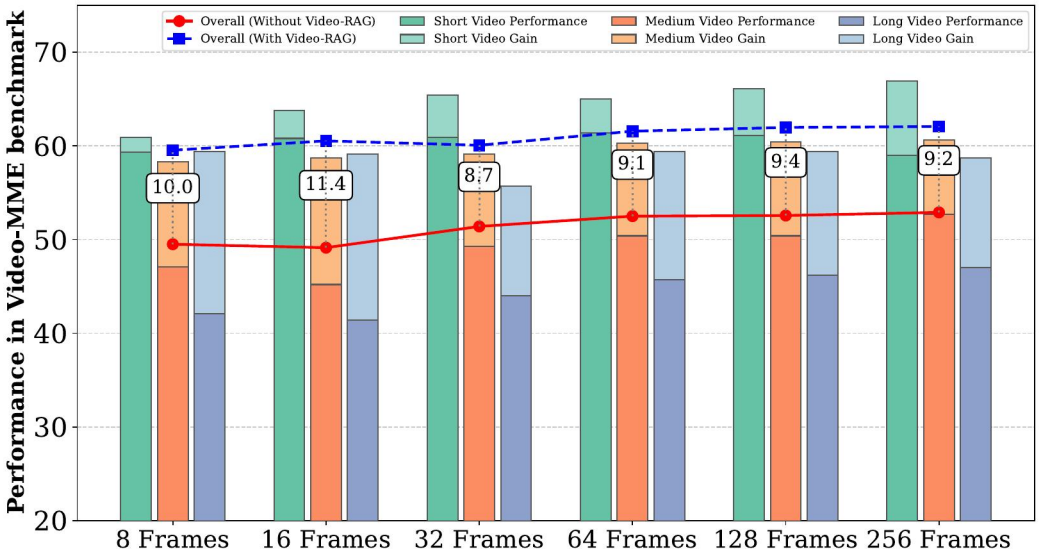

Figure 3: Performance gain with different frame sampling rates on Video-MME using LongVA-7B. Video-RAG shows consistent improvements across different frame rates, with larger gains at higher frame rates where redundancy is more pronounced.

- Object detection information (DET) combined with ASR provides the strongest improvements (+6.1% overall), particularly excelling on long videos.
- The complete Video-RAG system achieves the best performance (+7.1% overall) by leveraging all three modalities.

The results demonstrate that different auxiliary text types complement each other, with the full system consistently outperforming individual components.

Retrieval Threshold Analysis. Table 5 examines the effect of different retrieval thresholds on Video-MME using Long-LLaVA-7B. The analysis reveals an optimal threshold of $t = 0.3$, which achieves the best overall performance (62.1%) while maintaining reasonable token efficiency (1.9K tokens) and inference time (11s per question).

Lower thresholds (t=0.0-0.2) include more auxiliary text but may introduce noise, while higher thresholds (t=0.4-0.5) become too restrictive. The comparison with random text sampling (59.1% with 1.9K tokens) confirms that targeted retrieval is crucial for Video-RAG's effectiveness.

Frame Rate Analysis. Figure 3 shows Video-RAG's performance across different frame sampling rates using LongVA-7B on Video-MME. The results demonstrate that Video-RAG provides consistent

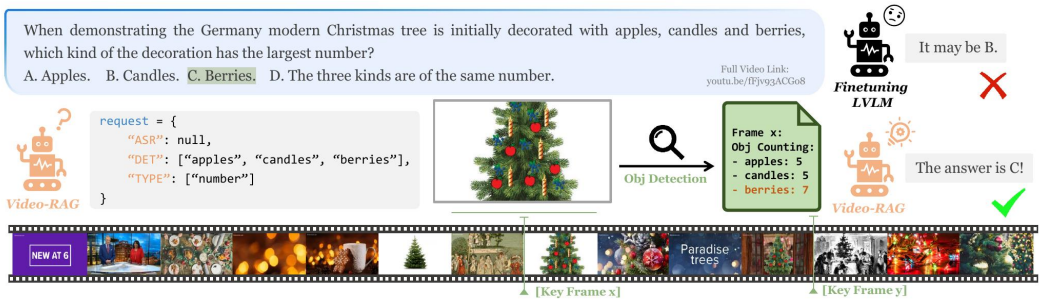

Figure 4: Qualitative example showing Video-RAG's effectiveness on Video-MME. The retrieved auxiliary texts provide crucial context about conversation content and object locations that enable correct answer selection.

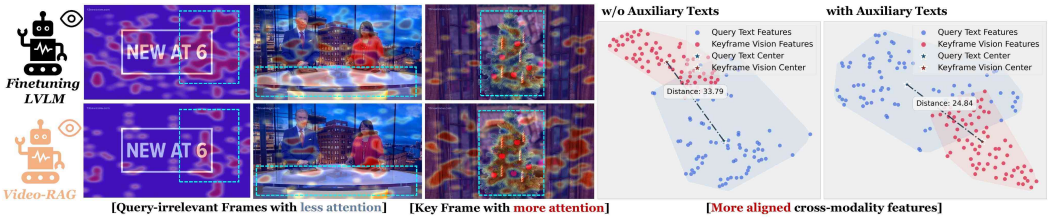

Figure 5: Grad-CAM visualizations and t-SNE analysis showing improved query-frame alignment with Video-RAG. The retrieved auxiliary texts help focus attention on relevant video regions and improve semantic alignment between query and visual content.

benefits across all frame rates, with particularly strong gains at higher frame rates where visual redundancy is more pronounced. This validates our core hypothesis that replacing redundant visual tokens with targeted auxiliary texts is an effective strategy.

4.4 Qualitative Analysis

Figure 4 presents a qualitative example demonstrating Video-RAG's effectiveness. The original query asks about the total number of people in a video conversation. While visual frames may not clearly show all participants, the retrieved ASR information provides dialogue content that reveals the presence of multiple speakers, and the OCR information captures additional textual cues. This multimodal evidence enables the model to correctly identify the answer.

Figure 5 provides visualization analysis using Grad-CAM heatmaps and t-SNE embeddings. The results show that Video-RAG improves query-frame alignment by providing semantic bridges between textual queries and visual content. The t-SNE visualization reveals that auxiliary texts create better semantic clusters, facilitating more accurate cross-modal reasoning.

4.5 Efficiency Analysis

Video-RAG introduces minimal computational overhead while providing substantial performance gains. On 7B models, the additional GPU memory requirement is approximately +8GB, and the average inference time increases by 5s per case. The token overhead averages 2.0K tokens, equivalent to 14 additional frames, but provides significantly more targeted information than random frame expansion.

The offline processing for auxiliary text extraction (OCR, ASR, object detection) can be performed once per video and cached for multiple queries, making the approach highly practical for real-world deployment. The retrieval process itself is efficient, requiring only embedding computations and similarity search operations.

Our threshold-based retrieval strategy ensures that only relevant auxiliary texts are included, maintaining a favorable balance between performance improvement and computational cost. The consistent

improvements across different model architectures and video lengths demonstrate the robustness and generalizability of the Video-RAG approach.

5 Conclusion

We present Video-RAG, a training-free, plug-and-play pipeline that enhances long video understanding by strategically incorporating query-relevant auxiliary texts extracted from video content. Our approach addresses the fundamental challenge of limited context windows in LVLMs by replacing redundant visual tokens with targeted textual information, enabling more efficient use of available context.

The key innovation of Video-RAG lies in its visually-aligned retrieval strategy, which combines OCR, ASR, and object detection information through sophisticated similarity search. By using only open-source tools—EasyOCR, Whisper-large, APE, CLIP, Contriever, and FAISS—our approach democratizes access to advanced video understanding capabilities without requiring proprietary models or expensive retraining.

Our comprehensive evaluation across Video-MME, MLVU, and LongVideoBench demonstrates consistent improvements across multiple model architectures and video lengths. The +2.8% average improvement on Video-MME with only 2.0K additional tokens highlights the efficiency of our approach. Most notably, enabling LLaVA-Video-72B to surpass GPT-4o performance (77.4% vs. 77.2%) represents a significant milestone in achieving proprietary-level capabilities with open-source solutions.

The ablation studies reveal that different auxiliary text modalities contribute complementarily, with the complete system consistently outperforming individual components. The optimal retrieval threshold of t=0.3 provides an effective balance between information richness and computational efficiency.

Limitations and Future Work. While Video-RAG shows strong performance improvements, several limitations remain. The approach currently focuses on English content, and extending to multilingual videos would require additional considerations. The auxiliary text extraction quality depends on the underlying tools (OCR, ASR, object detection), and improvements in these components would directly benefit Video-RAG performance.

Future work could explore adaptive threshold selection based on query complexity, integration with more sophisticated scene understanding models, and extension to other video-language tasks such as video generation and editing. Additionally, investigating the optimal combination strategies for different auxiliary text modalities could yield further improvements.

Video-RAG represents a significant step toward efficient, accessible long video understanding, demonstrating that strategic information retrieval can achieve substantial performance gains without the computational overhead of traditional approaches. The training-free nature and reliance on open-source tools make this approach immediately applicable to existing video understanding systems, potentially accelerating progress in the field.

6 Acknowledgments

We thank the open-source community for providing the foundational tools that make Video-RAG possible, including the teams behind EasyOCR, Whisper, CLIP, Contriever, and FAISS. We also acknowledge the creators of the Video-MME, MLVU, and LongVideoBench datasets for enabling comprehensive evaluation of video understanding systems. Special thanks to the LLaVA team for their continued contributions to open-source vision-language models.

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